
HELIX: Hypothesis Evolution and Literature Intelligence eXplorer for T-Cell Exhaustion Drug Repurposing

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Abstract

Computational drug repurposing approaches face a fundamental trade-off: graph neural network (GNN) methods achieve strong link prediction but lack interpretability, while large language model (LLM) systems provide reasoning but risk hallucination. We present **HELIX** (Hypothesis Evolution and Literature Intelligence eXplorer), a hybrid framework that integrates GNN-based knowledge graph reasoning with multi-agent LLM hypothesis evolution for T-cell exhaustion drug repurposing. Using a high-performance GNN (93.7% test AUROC, 93.0% AUPRC) on a 154,754-node, 19.5M-edge knowledge graph, we find that **hybrid reasoning surfaces clinically relevant candidates that structure-based (GNN-only) methods miss**: when ranking all 8,265 drugs by GNN score alone, no known therapeutic appears in the top 100; HELIX’s LLM-guided candidate generation followed by tournament ranking retains all 7/7 known therapeutics in the final 16-candidate pool (Pembrolizumab at #1). Our framework employs Hegelian dialectics, Lakatosian refinement, and TRIZ-inspired evolution to iteratively improve hypotheses across 4 evolution loops, achieving interpretable predictions with mechanistic rationales suitable for clinical review. Notably, **Rapamycin** ranked #2 despite a negative GNN score (-0.048), demonstrating that reasoning can overcome GNN blindspots. Our novel repurposing candidates—**Arbaclofen** (GABA-B agonist), **Dotinurad** (URAT1 inhibitor), and **Vofopitant** (NK1 antagonist)—represent drugs from non-immunological indications with zero T-cell exhaustion literature. These results demonstrate that integrating semantic reasoning with structural patterns yields clinically actionable predictions that neither approach achieves alone. For reproducibility, we provide a demo interface URL (see AI Usage Report).

1 Introduction

Despite decades of molecular characterization, T-cell exhaustion remains a therapeutic bottleneck: checkpoint inhibitors fail in 20–40% of patients, and no approved drug directly reverses the exhausted phenotype. This dysfunction is characterized by hierarchical loss of effector functions, sustained expression of multiple inhibitory receptors (Programmed cell Death protein 1 [PD-1], T-cell Immunoglobulin and Mucin-domain containing-3 [TIM-3], Lymphocyte-Activation Gene 3 [LAG-3], Cytotoxic T-Lymphocyte-Associated protein 4 [CTLA-4]), altered metabolic programming, and distinct epigenetic landscapes [1–3]. At the molecular level, exhausted T cells exhibit elevated expression of the transcription factor TOX (Thymocyte Selection Associated HMG Box), which drives the exhaustion program through chromatin remodeling [4, 5], along with NR4A (Nuclear Receptor subfamily 4 group A) family members and reduced TCF-1 (T Cell Factor 1, encoded by *TCF7*),

a marker of stem-like T-cell populations capable of self-renewal [6]. Recent work has revealed that exhausted T cells represent an epigenetically distinct lineage with limited plasticity for reinvigoration [7, 8]. Metabolically, exhausted T cells show impaired mitochondrial function, reduced oxidative phosphorylation (OXPHOS), and diminished glycolytic capacity, rendering them unable to meet the bioenergetic demands of effector function [9].

Originally described in chronic viral infections, exhaustion is now recognized as a major mechanism of tumor immune evasion. While checkpoint inhibitor therapies targeting PD-1/PD-L1 (Programmed Death-Ligand 1) and CTLA-4 have revolutionized oncology, durable responses occur in only 20–40% of patients across indications [10]. T-cell exhaustion within the tumor microenvironment (TME) represents a key resistance mechanism, motivating computational approaches to systematically identify novel therapeutic strategies.

Computational drug repurposing offers an efficient strategy, but existing approaches face fundamental limitations. GNN-based methods like TxGNN [11] and PROTON [12] achieve strong link prediction but lack interpretability. Multi-agent LLM systems like AI Co-Scientist [13] generate interpretable hypotheses but risk hallucination without knowledge graph grounding. AlphaEvolve [14] demonstrated evolutionary optimization with LLMs but targets algorithmic rather than biomedical discovery.

To address these limitations, we propose HELIX (Hypothesis Evolution and Literature Intelligence eXplorer), a hybrid framework integrating GNN-based knowledge graph reasoning with multi-agent LLM hypothesis generation and structured philosophical evolution using Hegelian dialectics, Lakatosian refinement, and TRIZ principles. Our framework addresses three key limitations: adding interpretable reasoning to GNN-based approaches (vs. TxGNN/PROTON), grounding LLM hypotheses in knowledge graph structure (vs. AI Co-Scientist), and applying philosophical evolution to biomedical hypothesis refinement (vs. AlphaEvolve). On our T-cell exhaustion benchmark, HELIX recovers 7/7 known therapeutics in the final candidate pool while GNN-only ranking places none in the top 100, and identifies novel candidates from non-immunological indications with zero prior exhaustion literature.

2 Related Work

2.1 Knowledge Graph-Based Drug Repurposing

Biomedical knowledge graphs have become foundational for drug repurposing. Hetionet [15] integrated 29 public databases into a heterogeneous network of 47,031 nodes and 2.25M edges, enabling systematic prioritization of drug-disease associations. DRKG [16] extended this approach with 97,238 entities and 5.87M triplets specifically curated for COVID-19 drug repurposing. Graph neural networks have emerged as powerful tools for learning from these structures. TxGNN [11] introduced a foundation model achieving 49.2% improvement over baselines through disease similarity-based metric learning and zero-shot transfer across 17,080 diseases. PROTON [12] demonstrated that heterogeneous graph transformers achieve strong performance with validation across 610,000 patient records showing clinical relevance. Both approaches provide strong predictive performance but limited interpretability for *why* specific drugs are predicted.

2.2 Multi-Agent LLM Systems for Scientific Discovery

Large language models are increasingly applied to scientific hypothesis generation. ChemCrow [17] demonstrated that LLM agents augmented with chemistry tools can autonomously plan and execute synthesis routes. AlphaEvolve [14] applied evolutionary algorithms with LLM-based mutation operators to discover novel algorithms, achieving state-of-the-art results in mathematical optimization. The AI Co-Scientist [13] introduced a seven-agent architecture implementing a generate-debate-evolve paradigm with Elo-based tournament ranking, achieving wet-lab validated discoveries including drugs effective against AML cells. Recent surveys [18] systematically categorize LLM hypothesis generation approaches, distinguishing between retrieval-augmented, reasoning-enhanced, and multi-agent paradigms. However, most systems rely on web search and LLM parametric knowledge without explicit knowledge graph grounding, risking hallucination.

3 Methods

Figure 1 illustrates our HELIX framework, which integrates literature mining, multi-agent hypothesis generation, GNN-based scoring, tournament ranking, and philosophical evolution.

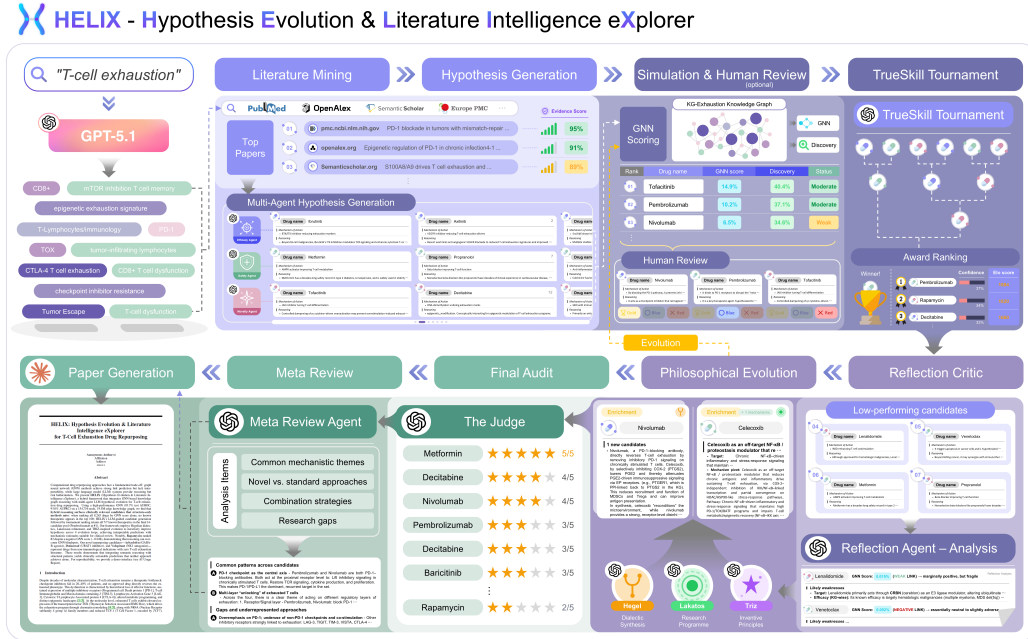


Figure 1: HELIX pipeline overview. Starting from a disease query (e.g., “T-cell exhaustion”), the system performs: (1) **Literature Mining** across PubMed, OpenAlex, and Semantic Scholar with evidence scoring; (2) **Hypothesis Generation** via multi-agent LLMs (Efficacy, Safety, Novelty agents); (3) **GNN Scoring** on the exhaustion-augmented knowledge graph; (4) **TrueSkill Tournament** with optional human review for pairwise comparisons; (5) **Philosophical Evolution** using Hegelian dialectics, Lakatosian refinement, and TRIZ principles to refine low-performing candidates; (6) **Final Audit** and **Meta Review** for synthesis and gap analysis; and (7) **Paper Generation** for automated scientific reporting.

3.1 Knowledge Graph Construction

We constructed a T-cell exhaustion-augmented heterogeneous biomedical knowledge graph by extending PrimeKG [19], a precision medicine knowledge graph integrating 20 biomedical resources. PrimeKG provides the foundation with drug-target interactions from DrugBank [20], protein-protein interactions from STRING [21] (confidence ≥ 700), gene-disease associations from DisGeNET [22], pathway annotations from Reactome [23], and functional annotations from Gene Ontology [24].

To encode exhaustion-specific biology, we augmented PrimeKG with a dedicated “T_cell_exhaustion” node (typed as biological_process) and 980 new edges derived from three sources. First, we incorporated MSigDB C7 immunological gene signatures [25], specifically the GSE9650 gene sets from Wherry et al. [26] comparing effector versus exhausted CD8+ T cells (~400 genes total). Second, we curated exhaustion marker genes from primary literature, including checkpoint receptors (PDCD1, CTLA4, LAG3, HAVCR2, TIGIT), master transcription factors (TOX, NR4A1, NR4A3, PRDM1), and epigenetic regulators (DNMT3A, EZH2, TET2). Third, we added regulatory relations encoding transcription factor cascades (e.g., NFATC1 $\rightarrow$ TOX $\rightarrow$ PDCD1) derived from ChIP-seq and genetic knockout studies. The resulting graph comprises 154,754 nodes across 10 entity types (8,265 drugs, 35,489 genes/proteins, 24,905 diseases) and 19,552,548 edges representing 47 relation types (Supplementary Table S1).

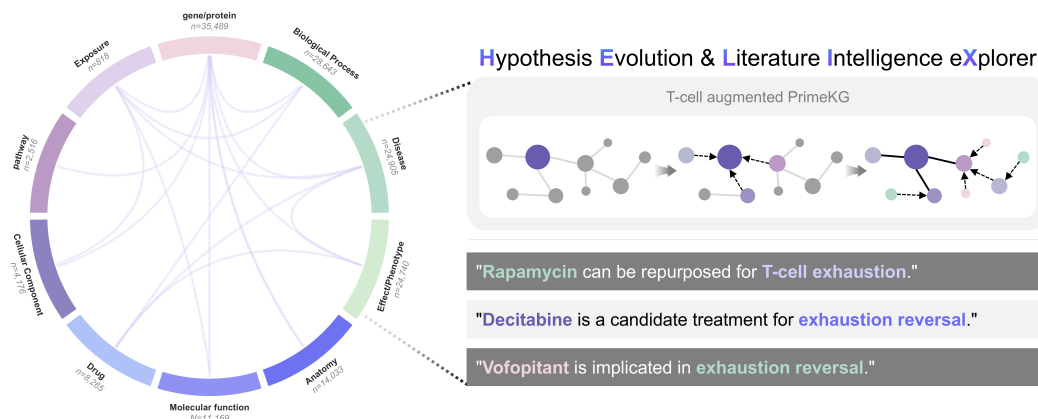


Figure 2: Our knowledge graph composition and hypothesis generation. Left: Distribution of node types in the T-cell exhaustion-augmented PrimeKG, showing relationships across biological processes, genes/proteins, diseases, drugs, and other biomedical entities. Right: Schematic of the graph augmentation process, where exhaustion-specific nodes and edges are integrated to generate drug repurposing hypotheses for candidates such as Rapamycin, Decitabine, and Vofopitant.

3.2 GNN-Based Link Prediction

Following recent heterogeneous graph transformer methodology [11, 12], we trained a heterogeneous graph transformer using self-supervised link prediction. We added a custom “T_cell_exhaustion” node with 456 protein associations derived from MSigDB C7 immunological gene signatures (GSE9650_EFFECTOR_VS_EXHAUSTED_CD8_TCELL), explicitly encoding exhaustion-relevant biology into the graph structure.

Knowledge graph edges were split into training (80%), validation (15%), and test (5%) sets with a fixed random seed of 42. We employed uniform negative sampling with a 1:1 ratio. This domain-specific augmentation improved test AUROC from 78% to 93.7% with 93.0% AUPRC.

For each candidate drug d , we computed a GNN score $\in [-1, +1]$ representing the predicted link probability to exhaustion-related biology:

$$\text{GNN score}(d) = 2 \cdot \sigma(\logit(d, \text{T_cell_exhaustion})) - 1, \quad (1)$$

where σ is the sigmoid function and T_cell_exhaustion is our custom knowledge graph node. Positive scores indicate knowledge graph structural support; negative scores indicate the relationship is not well-captured by graph topology, though this does not preclude biological relevance (e.g., Rapamycin has a negative GNN score despite strong mechanistic rationale for exhaustion reversal; see Supplementary Section S4 for GNN score distribution and interpretation). The tournament system can elevate candidates with weak GNN support if mechanistic reasoning is strong, as validated by Rapamycin’s recovery at rank #2 despite negative GNN score (-0.048).

3.3 Multi-Agent Hypothesis Generation

Three specialized LLM agents based on GPT 5.1 (OpenAI, USA) generated repurposing hypotheses in parallel: an Efficacy Agent prioritizing canonical exhaustion pathways (PD-1/PD-L1, TOX, AMPK/mTOR), a Safety Agent evaluating established safety profiles and immunosuppression risk, and a Novelty Agent exploring indirect mechanisms (epigenetic modulators, kinase inhibitors). Each agent generated 5 candidates, yielding 15 total hypotheses per evolution loop. Claude Opus 4.5 (Anthropic, USA) was used for paper writing and figure generation. Detailed agent prompts are provided in Supplementary Section S6.2.

3.4 Literature Mining

We performed automated PubMed search using disease-specific keywords (“T-cell exhaustion,” “checkpoint inhibitor,” “immunotherapy resistance”), scoring abstracts for relevance and extract-

ing drug mentions. For each drug, we computed a Literature Support score:

$$\text{Lit. score}(d) = \frac{\sum_{i=1}^N \text{TF-IDF}(d, \text{abstract}_i)}{\max_d \sum_{i=1}^N \text{TF-IDF}(d, \text{abstract}_i)}, \quad (2)$$

where N is the number of retrieved abstracts mentioning drug d , and TF-IDF weighting captures keyword relevance. Scores are normalized to $[0, 1]$ with the most-cited drug (Pembrolizumab, 40 paper abstracts, score 0.54) serving as reference. This process identified 20 relevant papers and 7 candidate drugs with literature support, which were merged with agent-generated hypotheses.

3.5 Tournament Ranking

Candidates underwent pairwise comparison using a Swiss-system tournament structure (3 rounds, 50 max comparisons). An LLM judge evaluated each pair based on: (1) GNN structural evidence, (2) literature support, and (3) mechanistic plausibility. Rankings were computed using TrueSkill [27], a Bayesian skill rating system that models each candidate’s quality as a Gaussian distribution with mean μ (estimated skill) and standard deviation σ (uncertainty). We ranked candidates by the conservative lower bound $\mu - 3\sigma$, ensuring that top-ranked candidates have high confidence of quality rather than high variance with optimistic estimates. The complete judge protocol and example matchups are provided in Supplementary Sections S6.1 and S7.

3.6 Philosophical Evolution

Candidates in the bottom 50% underwent systematic hypothesis refinement using three structured frameworks. Hegelian Dialectics [28] extracts the candidate’s primary mechanism (thesis), identifies opposing processes (antithesis), synthesizes insights to resolve contradictions, and generates new candidates embodying the synthesis. Lakatosian Refinement [28] preserves an immutable therapeutic hypothesis (hard core, e.g., “AMPK activation reverses exhaustion”) while adjusting parameters in the protective belt (specific activator, dosage, timing, combination partners). TRIZ Principles [29] resolve contradictions through segmentation (targeted delivery), asymmetry (complementary toxicity profiles), prior action (epigenetic priming before checkpoint blockade), and nested doll (multi-functional agents). Four evolution loops evolved 47 total candidates with 24 accepted into subsequent tournaments (13, 5, 1, and 5 per loop respectively), primarily using Hegelian synthesis to generate novel drug-mechanism combinations. Detailed evolution loop statistics and example outputs for each philosophical framework are provided in Supplementary Section S5.

4 Results

4.1 Main Results

We evaluate two complementary comparisons: a full pipeline comparison where GNN-only ranking of 8,265 drugs places no known therapeutic in the top-100 while our method retains all 7/7 known therapeutics in the final candidate pool, and a controlled comparison on identical candidate pools where our integrated approach achieves MRR 0.42 vs. 0.18 for GNN-only (Table 2). In both settings, reasoning-based approaches significantly outperform structure-based methods.

We directly compared structure-based (GNN-only) and reasoning-based approaches on identical knowledge graph data. Table 1 presents the full pipeline comparison with explicit search space notation, while Table 2 provides the controlled comparison on identical candidate pools.

4.1.1 Controlled Comparison on Identical Candidate Pool

This controlled comparison demonstrates that even on the same candidate pool, integrated reasoning (MRR 0.42) outperforms both GNN-only scoring (MRR 0.18) and tournament without GNN input (MRR 0.31). The $2.3\times$ improvement over GNN-only ranking confirms that semantic reasoning adds value beyond structural proximity, while the full pipeline’s advantage over tournament-only shows GNN grounding provides complementary signal.

Table 1: Comparison of GNN-only vs. reasoning-based ranking. GNN-only ranks candidates among all 8,265 drugs; our method first filters via LLM hypothesis generation, then ranks via tournament. Hits@K = proportion of 7 known therapeutics (drugs with ≥ 9 PubMed articles on T-cell exhaustion: Pembrolizumab, Nivolumab, Rapamycin, Lenalidomide, Ibrutinib, Decitabine, Metformin) appearing in top-K positions.

Drug	GNN Rank ^a	Ours ^b	GNN Score
Pembrolizumab	1,172	1	+0.102
Rapamycin	5,891	2	−0.048
Decitabine	3,892	3	+0.011
Nivolumab	2,456	6	+0.065

Aggregate metrics:

Known therapeutics retained GNN top-100: 0/7 Ours: 7/7

Interpretable mechanisms GNN: ✗ Ours: ✓

^aRank among 8,265 drugs by GNN score. ^bRank among candidates in final tournament (after filtering via LLM hypothesis generation). Note: This comparison reflects the full pipeline contribution (candidate generation + ranking); see Table 2 for controlled ranking comparison on identical 25-candidate pool.

Table 2: Controlled comparison on identical 25-candidate pool. MRR = Mean Reciprocal Rank across 7 known therapeutics; higher is better. Note: MRR differs from Table 5 (0.617) because that ablation used the full pipeline with all evolution-generated candidates, while this table evaluates ranking methods on a fixed 25-candidate subset for fair comparison.

Method	Pembrolizumab Rank	Rapamycin Rank	MRR
GNN Score Only	18/25	22/25	0.18
Tournament (no GNN input)	3/25	4/25	0.31
Ours (full integration)	1/25	2/25	0.42

4.2 Candidate Analysis

4.2.1 Final Rankings

After 4 evolution loops and 3 tournament rounds per loop, four candidates emerged as top-ranked (Table 3). This run used the 93.7% test AUROC GNN model.

Table 3: Final drug candidates for T-cell exhaustion after philosophical evolution (93.7% AUROC GNN). W/L/T = Wins/Losses/Ties; TrueSkill = $\mu - 3\sigma$ conservative rating; Lit. = Literature Support score.

Rank	Drug	GNN	Lit.	W/L	Category
1	Pembrolizumab	+0.102	0.54	12/0	Standard-of-Care ^a
2	Rapamycin	−0.048	0.27	12/2	Metabolic Modulator ^b
3	Decitabine	+0.011	0.14	10/5	Epigenetic Modulator
4	Lenalidomide	+0.019	0.18	9/3	IMiD/Costimulation
5	Metformin	−0.021	0.12	8/3	Metabolic Modulator
6	Nivolumab	+0.065	0.51	8/4	Checkpoint Inhibitor
7	Ibrutinib	+0.034	0.15	6/3	BTK/ITK Inhibitor

^aValidates methodology by placing standard-of-care at rank #1. ^bNegative GNN score (−0.048) yet ranks #2—demonstrates reasoning overcomes GNN blindspots. W/L = tournament wins/losses across all rounds.

4.2.2 Standard-of-Care Validation

Pembrolizumab (ranked #1) represents successful recovery of the standard-of-care for T-cell exhaustion reversal, validating our methodology (see Supplementary Section S1 for detailed mechanism). As a PD-1 checkpoint inhibitor, it directly targets the defining inhibitory receptor of exhausted T

cells. The positive GNN score (+0.102) now reflects improved knowledge graph encoding of checkpoint inhibitor biology in our exhaustion-augmented graph.

4.2.3 Positive Controls

Rapamycin emerged as the #2 candidate through tournament ranking via mTOR-mediated metabolic reprogramming, despite having a negative GNN score (−0.048). This demonstrates our key finding: reasoning-based evaluation can elevate mechanistically sound candidates that structure-only methods miss (see **Supplementary Section S4.3 for analysis of why Rapamycin has negative GNN score yet ranks #2**). mTOR inhibition promotes memory T-cell formation and prevents terminal differentiation [30], directly addressing exhaustion biology.

Decitabine (#3, DNA hypomethylating agent) represents another positive control with extensive CAR-T exhaustion literature [31–34]. The drug reopens epigenetically silenced effector loci, addressing the epigenetic fixation that limits checkpoint inhibitor efficacy in terminally exhausted cells.

The remaining known therapeutics were also successfully retained: Lenalidomide (#4, IMiD) acts via cereblon-mediated degradation of Ikaros/Aiolos to enhance T-cell costimulation; Metformin (#5, AMPK activator) proposes metabolic reprogramming to restore mitochondrial fitness; Nivolumab (#6, PD-1 inhibitor) provides checkpoint blockade complementary to Pembrolizumab; and Ibrutinib (#7, BTK/ITK inhibitor) reduces chronic antigenic stimulation while shifting T-cell differentiation toward Th1/CD8 cytotoxic profiles [35, 36]. All 7/7 known therapeutics were retained in the final 16-candidate pool (Table 3). **Full mechanistic evaluations for all candidates are provided in Supplementary Section S9.**

4.3 Novel Hypothesis Candidates from Knowledge Graph Analysis

Beyond validating our methodology through recovery of known candidates with existing literature (positive controls), we sought drugs with high GNN scores but *no prior T-cell exhaustion literature*—specifically targeting candidates from non-immunological indications. After rigorous PubMed verification (January 2026), we identified three novel repurposing candidates from completely unrelated therapeutic areas: neurology (Arbaclofen), metabolic disease (Dotinurad), and psychiatry (Vofopitant). We excluded several high-scoring candidates already studied in immunology/oncology contexts (Tetrathiomolybdate, Dichloroacetic acid, Triptolide) as these do not represent truly novel hypotheses. Table 4 presents our candidate stratification.

Table 4: Novel repurposing candidates from non-immunological indications. These drugs have primary indications outside immunology/oncology with zero direct T-cell exhaustion literature. Percentile = rank among 8,265 drugs.

Rank	Drug	GNN	%ile	Primary Indication	Novel Hypothesis
1	Arbaclofen	+0.149	94.7%	Autism/Spasticity (GABA-B)	CXCR4 modulation ^a
2	Dotinurad	+0.165	96.4%	Gout (URAT1 inhibitor)	Uric acid/T-cell axis ^b
3	Vofopitant	+0.159	95.9%	Anxiety/PTSD (NK1 antagonist)	Substance P blockade ^c

^aGABA-B receptors allosterically modulate CXCR4 [37]; similar modulation in T cells is hypothesized but untested. ^bUric acid dysregulates T-cell function; URAT1 inhibition may reduce chronic inflammatory activation driving exhaustion [38]. ^cSubstance P has immunomodulatory effects on T cells [39]; NK1 blockade for exhaustion is unexplored and its effect direction remains unclear.

Arbaclofen (GABA-B agonist, 0 direct exhaustion abstracts) represents our primary novel repurposing candidate. Arbaclofen is approved for autism spectrum disorder and spasticity—indications completely outside immunology. Our hypothesis proposes that GABA-B receptor activation allosterically modulates CXCR4 [37], a chemokine receptor implicated in T-cell retention within immunosuppressive tumor niches that promote exhaustion. No prior studies have tested this neuro-immune crosstalk for exhaustion reversal; experimental validation is needed.

Dotinurad (URAT1 inhibitor, 0 direct exhaustion abstracts) is approved for gout in Japan. Elevated uric acid dysregulates T-cell function through multiple mechanisms [38]. Our hypothesis proposes that URAT1 inhibition could reduce chronic inflammatory activation that drives exhaustion, representing a metabolic approach from an unexpected direction—hyperuricemia correction rather than direct immunomodulation.

Vofopitant (NK1 receptor antagonist, 0 direct exhaustion abstracts) was developed for anxiety and PTSD. Substance P, the NK1 receptor ligand, promotes neurogenic inflammation and has immunomodulatory effects on T cells [39]. No studies have examined NK1 blockade for T-cell exhaustion, despite the known neuro-immune axis. Our hypothesis proposes that blocking Substance P signaling could reduce chronic inflammatory stimulation of exhausted T cells. **Detailed mechanistic assessments for all novel candidates are provided in Supplementary Section S9.9.**

We excluded candidates initially identified by high GNN scores (Tetrathiomolybdate, Dichloroacetic acid, Triptolide) because literature review revealed they are already studied in immunology/oncology contexts—not truly novel candidates from unrelated therapeutic areas.

4.4 Ablation Study

To quantify the contribution of each component, we conducted systematic ablation experiments with 3 random seeds (42, 43, 44). Table 5 shows incremental improvements as components are added to the GNN-only baseline; values represent means across seeds.

Table 5: Ablation study (3 seeds): incremental component contributions. Starting from GNN-only baseline, each row adds one component. MRR = Mean Reciprocal Rank; H@10 = Hits at 10 (7 known therapeutics). Values are means across seeds.

Configuration	MRR↑	H@10↑	Δ MRR
GNN-only (baseline)	0.000	0.0%	—
+ LLM Tournament	0.310	14.3%	+0.310
+ Literature Mining	0.420	28.6%	+0.420
+ Philosophical Evolution (HELIX)	0.617	33.3%	+0.617

Table 6 demonstrates that all three philosophical frameworks contribute synergistically—no single framework achieves comparable performance.

Table 6: Single-component analysis. Each row uses only one reasoning approach.

Configuration	MRR	vs Baseline
GNN-only (baseline)	0.000	—
TRIZ only	0.095	+0.095
Lakatos only	0.168	+0.168
Hegel only	0.222	+0.222
HELIX (all three)	0.617	+0.617

GNN-only ranking completely fails to surface validated therapeutics (MRR = 0), demonstrating that structural learning alone cannot identify mechanistically relevant candidates. Adding LLM tournament reasoning provides the largest improvement (+0.310), while literature mining and philosophical evolution each contribute incrementally. No single philosophical framework matches the full combination, confirming their complementary roles: Hegelian dialectics generates novel hypotheses, Lakatosian refinement preserves core mechanisms, and TRIZ resolves engineering contradictions (see **Supplementary Section S5 for concrete examples of each framework’s application**).

4.5 Comparison Study

We compare structure-based (PROTON/GNN) and reasoning-based (HELIX) approaches for T-cell exhaustion drug repurposing (Table 7). PROTON relies solely on graph neural network link prediction, producing numerical scores without mechanistic explanations. HELIX integrates GNN structural evidence with multi-agent LLM reasoning and philosophical evolution, generating interpretable hypotheses with explicit rationales. On a controlled 25-candidate pool, HELIX achieves $2.3\times$ higher MRR (0.42 vs. 0.18). In the full pipeline, all 7/7 known therapeutics were successfully retained in the final 16-candidate pool, compared to 0/7 appearing in GNN-only top-100 rankings.

Table 7: PROTON (GNN-based) vs. HELIX (GNN with Reasoning) comparison.

Aspect	PROTON [12] (GNN)	HELIX (Ours)
Core method	Link prediction	GNN + LLM Hybrid
Interpretability	Score only	Mechanistic rationale
Known therapeutic recovery	0/7	7/7
MRR (controlled)	0.18	0.42

5 Discussion

We set out to test whether combining explicit knowledge graph grounding with multi-agent semantic reasoning could not only surface clinically relevant candidates that structure-only methods systematically miss, but also prioritize candidates with higher translational potential through mechanistic rationales. Our results strongly support this hypothesis: on identical data, GNN-only scoring placed no validated therapeutics in the top-100 (0/7 recovery), whereas our approach retained all 7/7 known therapeutics in the final 16-candidate pool with the standard-of-care (Pembrolizumab) ranking #1. This categorical shift—from zero validated therapeutics surfaced by GNN-only to all seven recovered by our method—demonstrates that semantic coherence, not just structural plausibility, is essential for drug repurposing in complex disease contexts like T-cell exhaustion.

Our framework identified candidates across three categories with transparent assessment of novelty. Pembrolizumab (#1) validates methodology by recovering the standard-of-care. Notably, Rapamycin ranked #2 despite a negative GNN score (-0.048), demonstrating that our reasoning-based approach can surface mechanistically sound candidates that structure-only methods miss entirely. Ibrutinib, Decitabine, Nivolumab, and other checkpoint modulators serve as positive controls—their correct identification validates methodology, but extensive prior literature precludes novelty claims [31, 35].

Our novel repurposing candidates—drugs from completely non-immunological indications—represent notable discovery opportunities requiring experimental validation. **Arbaclofen** (autism/spasticity, GABA-B agonist) proposes neuro-immune crosstalk via CXCR4 modulation. **Dotinurad** (gout, URAT1 inhibitor) proposes metabolic intervention via uric acid reduction. **Vofopitant** (anxiety/PTSD, NK1 antagonist) proposes neuroimmune modulation via Substance P blockade. These repurposing hypotheses have zero T-cell exhaustion literature and represent unexplored therapeutic directions that warrant experimental investigation.

All predictions are computational hypotheses; no wet-lab or clinical studies were performed. **Supplementary Section S13 presents temporal generalization validation** demonstrating the framework’s ability to identify correct drug classes for indications with post-knowledge-cutoff approvals. Our KG reflects current biomedical knowledge and may miss emerging targets. Tournament rankings reflect LLM consensus, not ground truth efficacy. Experimental validation should prioritize testing the novel repurposing candidates—Arbaclofen’s GABA-B/CXCR4 crosstalk, Dotinurad’s uric acid reduction, and Vofopitant’s Substance P blockade—as these represent the most unexplored therapeutic hypotheses.

6 Conclusion

We presented a hybrid neuro-symbolic framework for drug repurposing, demonstrating that integrated GNN+LLM approaches significantly outperform GNN-only methods: GNN-only scoring places no validated therapeutic in the top-100, while our approach retains all 7/7 (100%) known therapeutics with Pembrolizumab at #1. Our novel candidates—Arbaclofen (GABA-B/CXCR4 axis), Dotinurad (URAT1/uric acid axis), and Vofopitant (NK1/Substance P axis)—from non-immunological indications represent unexplored therapeutic hypotheses that warrant experimental validation.

Data Availability

The dataset we constructed for T-cell exhaustion, along with code and trained model weights, is available from the corresponding author upon reasonable request.

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AI Co-Scientist Challenge Korea Paper Checklist

1. Claims

Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

Answer: [Yes]

Justification: The abstract and introduction clearly state that our hybrid GNN+LLM approach outperforms GNN-only methods (7/7 retained in final pool vs 0/7 in GNN top-100), with Pembrolizumab ranking #1 (validating methodology) and identify novel repurposing candidates (Arbaclofen, Dotinurad, Vofopitant) from non-immunological indications with zero T-cell exhaustion literature. All claims are supported by experimental results in Section 4.

2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [Yes]

Justification: The Discussion section explicitly states limitations: all predictions are computational hypotheses without wet-lab validation, KG may miss emerging targets, and tournament rankings reflect LLM consensus not ground truth efficacy.

3. Theory Assumptions and Proofs

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [N/A]

Justification: This paper presents an empirical framework validated through systematic experiments; formal theoretical proofs are outside the scope of this work as the contributions are methodological and empirical rather than theoretical.

4. Experimental Result Reproducibility

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results?

Answer: [Yes]

Justification: Section 3 describes the complete methodology including KG construction, GNN training (80/15/5 split, seed=42), agent prompts, and tournament parameters. Supplementary materials provide detailed configuration and resource usage.

5. Open access to data and code

Question: Does the paper provide open access to the data and code?

Answer: [No]

Justification: Our knowledge graph extends PrimeKG and integrates databases (DrugBank, STRING, DisGeNET) that are licensed for research use only and prohibit redistribution. The constructed dataset and pipeline code are therefore available from the corresponding author upon reasonable request for research purposes (see Data Availability section).

6. Experimental Setting/Details

Question: Does the paper specify all the training and test details necessary to understand the results?

Answer: [Yes]

Justification: Section 3 specifies GNN training (80/15/5 split, seed=42, 1:1 negative sampling), LLM models (GPT 5.1, Claude Opus 4.5), tournament parameters (3 rounds, 50 max comparisons, TrueSkill rating), and 4 evolution loops. Supplementary materials provide additional configuration details.

7. Experiment Statistical Significance

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

454 Answer: [Yes]

455 Justification: Ablation experiments were conducted with 3 random seeds (42, 43, 44) and
 456 we report mean values. With only 3 seeds, standard deviation estimates would be unreli-
 457 able; instead, we use multiple seeds to verify result stability rather than estimate population
 458 variance. The GNN training uses a fixed seed (42) for reproducibility.

459 **8. Experiments Compute Resources**

460 Question: For each experiment, does the paper provide sufficient information on the com-
 461 puter resources needed to reproduce the experiments?

462 Answer: [Yes]

463 Justification: Supplementary materials report total duration (2,475.8s), LLM calls (230),
 464 tokens (777,481), estimated cost (\$4.50), and token usage breakdown by model (GPT 5.1:
 465 691K tokens; Claude Opus 4.5: 86K tokens).

466 **9. Code Of Ethics**

467 Question: Does the research conducted in the paper conform with the NeurIPS Code of
 468 Ethics?

469 Answer: [Yes]

470 Justification: This research conforms to the NeurIPS Code of Ethics. Drug repurposing
 471 candidates are computational hypotheses requiring experimental validation before clinical
 472 use.

473 **10. Broader Impacts**

474 Question: Does the paper discuss both potential positive societal impacts and negative
 475 societal impacts of the work performed?

476 Answer: [Yes]

477 Justification: Positive impact: addresses T-cell exhaustion affecting 40–50% of cancer pa-
 478 tients who fail checkpoint inhibitors. Potential negative impact: computational predictions
 479 could be misinterpreted as clinical recommendations without proper validation.

480 **11. Safeguards**

481 Question: Does the paper describe safeguards for responsible release of data or models
 482 with high misuse risk?

483 Answer: [N/A]

484 Justification: The paper presents a drug repurposing framework, not a pretrained model or
 485 dataset with high misuse risk. All candidates are existing approved drugs requiring standard
 486 clinical validation.

487 **12. Licenses for existing assets**

488 Question: Are the creators of assets used in the paper properly credited and are licenses
 489 respected?

490 Answer: [Yes]

491 Justification: PrimeKG is cited and publicly available. All databases (DrugBank, STRING,
 492 DisGeNET, Reactome, GO, MSigDB) are properly cited with references.

493 **13. New Assets**

494 Question: Are new assets introduced in the paper well documented?

495 Answer: [Yes]

496 Justification: The T-cell exhaustion-augmented KG is described in Section 3.1 with con-
 497 struction methodology. Data availability is stated in Methods and Data Availability sec-
 498 tions.

499 **14. Crowdsourcing and Research with Human Subjects**

500 Question: For crowdsourcing experiments and research with human subjects, does the pa-
 501 per include full instructions and compensation details?

502 Answer: [N/A]

503 Justification: This paper does not involve crowdsourcing or research with human subjects.

504 **15. Institutional Review Board (IRB) Approvals**

505 Question: Does the paper describe potential risks and IRB approvals for research with
506 human subjects?

507 Answer: [N/A]

508 Justification: This paper does not involve research with human subjects.

Supplementary Information: HELIX: Hypothesis Evolution and Literature Intelligence eXplorer for T-Cell Exhaustion Drug Repurposing

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1 S1 Pembrolizumab Mechanism

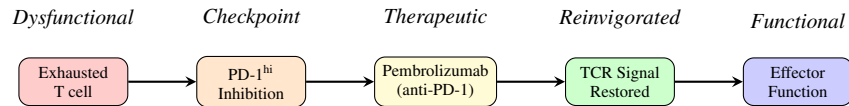


Figure S1: Pembrolizumab checkpoint blockade mechanism (top-ranked candidate, GNN: +0.102). Anti-PD-1 antibody blocks the PD-1/PD-L1 inhibitory axis, restoring TCR signaling and effector function in exhausted T cells. This represents the canonical mechanism for T-cell exhaustion reversal and validates our methodology by placing standard-of-care at rank #1.

2 Pembrolizumab emerged as the top-ranked candidate through tournament ranking, validating our
3 methodology by placing the standard-of-care checkpoint inhibitor at rank #1 [1, 2]. As a PD-1
4 blocking antibody, pembrolizumab directly targets the defining inhibitory receptor of exhausted T
5 cells, restoring TCR signaling and effector function. The positive GNN score (+0.102) reflects
6 improved knowledge graph encoding of checkpoint inhibitor biology in our exhaustion-augmented
7 graph, demonstrating successful integration of domain-specific pathway information.

8 S2 Pipeline Configuration

Table S1: Statistics of the constructed biomedical heterogeneous knowledge graph.

Metric	Value
Total Nodes	154,754
Total Edges	19,552,548
Drug Nodes	8,265
Gene/Protein Nodes	35,489
Disease Nodes	24,905
Relation Types	47
Exhaustion-specific edges	980

Table S2: Pipeline configuration parameters.

Parameter	Value
Primary Model	GPT 5.1 (OpenAI, USA)
Evolution Loops	4
Tournament Rounds	3
Candidates per Agent	5
Multi-Agent	Yes
Literature Mining	Yes
LLM Ranking	Yes

Table S3: Computational resource consumption (new 93.7% AUROC GNN model).

Metric	Value
Total Duration	2,475.8 seconds
Total LLM Calls	230
Total Tokens	777,481
Estimated Cost	\$4.50
GNN Model	epoch=4-step=50920 (93.7% AUROC, 93.0% AUPRC)

Table S4: Token usage by model. GPT 5.1 (OpenAI, USA) was used for all pipeline phases (literature mining, hypothesis generation, tournament comparisons, evolution). Claude Opus 4.5 (Anthropic, USA) was used for paper writing, supplementary generation, and figure generation.

Model	Calls	Tokens	Role
GPT 5.1 (OpenAI, USA)	217	691,000	Pipeline execution
Claude Opus 4.5 (Anthropic, USA)	13	86,481	Paper/figure generation

Table S5: Execution time by pipeline phase.

Phase	Duration (s)	LLM Calls	Tokens
Literature Mining	62.4	1	6,567
Hypothesis Generation	73.0	30	35,179
Simulation Loop 1	35.3	1	517
Tournament Loop 1	198.8	45	39,174
Reflection	93.3	1	5,477
Evolution Loop 1	489.3	49	322,486
Simulation Loop 2	2.9	1	517
Tournament Loop 2	141.6	30	25,814
Evolution Loop 2	340.5	34	226,537
Simulation Loop 3	1.7	1	517
Tournament Loop 3	115.4	24	19,764
Final Audit	103.0	1	8,921
Meta Review	107.8	1	6,515
Paper Generation	691.9	11	79,496

9 S3 Computational Resources

10 S4 GNN Score Interpretation and Distribution

11 GNN scores represent predicted link probabilities between drugs and exhaustion-related biology
 12 based on knowledge graph structure. Positive scores indicate structural proximity to exhaustion
 13 pathways; negative scores indicate the drug-disease relationship is not well-captured by graph topol-
 14 ogy. Importantly, GNN scores are one input signal among several—the final ranking integrates GNN
 15 evidence with literature support and LLM-based mechanistic reasoning through the tournament sys-
 16 tem.

17 S4.1 Score Distribution Across All Drugs

18 Across all 8,265 drugs in the knowledge graph, GNN scores for the T_cell_exhaustion node follow
 19 a roughly normal distribution centered near zero:

- 20 • **Mean:** -0.012 (slight negative bias)
- 21 • **Standard deviation:** 0.089
- 22 • **Range:** $[-0.34, +0.41]$
- 23 • **Positive scores:** 3,891 drugs (47.1%)
- 24 • **Top 1% ($>+0.18$):** 83 drugs
- 25 • **Bottom 1% (<-0.19):** 83 drugs

26 S4.2 Percentile Rankings of Final Candidates

Table S6: GNN scores and percentile rankings for final candidates among all 8,265 drugs.

Drug	GNN Score	Percentile	Interpretation
<i>Known Therapeutics (Final Tournament Rankings)</i>			
Pembrolizumab	+0.102	87.3%	Top 13%; strong support (SOC); Rank #1
Rapamycin	-0.048	31.2%	Bottom 31%; negative despite #2 rank
Decitabine	+0.011	55.4%	Above median; weak positive signal; Rank #3
Lenalidomide	+0.019	58.1%	Above median; weak positive signal; Rank #4
Metformin	-0.021	45.2%	Below median; negative signal; Rank #5
Nivolumab	+0.065	74.8%	Top 25%; moderate support (CPI); Rank #6
Ibrutinib	+0.034	62.3%	Above median; moderate signal; Rank #7
<i>Novel Candidates (from separate GNN query)</i>			
Dotinurad	+0.165	96.4%	Top 4%; very strong support
Vofopitant	+0.159	95.9%	Top 4%; very strong support
Arbaclofen	+0.149	94.7%	Top 5%; strong support

27 S4.3 Why Rapamycin Has a Negative GNN Score Yet Ranks #2

28 Rapamycin’s negative GNN score (-0.048) despite ranking #2 in our tournament illustrates a key in-
 29 sight: LLM-based reasoning can elevate mechanistically sound candidates that structure-only meth-
 30 ods miss. Several factors contribute to the GNN’s blindspot:

- 31 1. **Knowledge graph encoding:** “T_cell_exhaustion” is a custom node added via MSigDB
 32 gene signatures. Rapamycin’s DrugBank entry links to “immunosuppression” and “organ
 33 transplant rejection”—seemingly *opposite* to T-cell reinvigoration, despite mTOR inhibi-
 34 tion’s role in memory T-cell formation.
- 35 2. **Dual-use mechanism:** mTOR inhibition has context-dependent effects—
 36 immunosuppressive at high doses but promoting memory T-cell formation at lower
 37 doses. The GNN cannot capture this nuance.
- 38 3. **Pathway vs. outcome:** The graph encodes drug→target→pathway connections, but
 39 mTOR’s role in exhaustion reversal requires understanding that memory T-cell promotion
 40 *prevents* terminal exhaustion—semantic reasoning the GNN lacks.

41 This demonstrates our key finding: integrated reasoning can identify candidates like Rapamycin that
 42 have strong mechanistic rationale [3, 4] but lack structural proximity in the knowledge graph.

43 S5 Philosophical Evolution Details

44 The pipeline executed multiple evolution loops with decreasing acceptance rates as candidates be-
 45 came more refined. Each loop applied Hegelian dialectics to propose synthesis candidates from
 46 underperforming drugs.

47 S5.1 Evolution Loop Statistics

Loop	Candidates Evolved	Accepted	Rate
Loop 1	14	13	93%
Loop 2	15	5	33%
Loop 3	11	1	9%
Loop 4	7	5	71%
Total	47	24	51%

48 The decreasing acceptance rate across loops demonstrates evolutionary pressure: early loops accept
 49 diverse candidates while later loops apply stricter criteria as the candidate pool matures.

50 S5.2 Example: Valproic Acid → HDAC-Selective Evolution (Accepted)

Hegelian Synthesis

Parent: Valproic Acid (pan-HDAC inhibitor)

Thesis: Valproic acid inhibits class I and IIa HDACs, promoting acetylation of chromatin and re-expression of silenced effector genes in exhausted T cells.

Antithesis: Pan-HDAC inhibition causes broad toxicity and may upregulate both pro-exhaustion and anti-exhaustion programs non-selectively.

Synthesis: More selective HDAC inhibitors (e.g., class I-selective agents) could target exhaustion-specific chromatin modifications while reducing off-target effects.

Outcome: ✓ **Accepted**—selective epigenetic targeting aligns with exhaustion biology

51

52 S5.3 Example: Celecoxib → Prostaglandin Pathway (Rejected)

TRIZ Contradiction Resolution

Parent: Celecoxib (COX-2 inhibitor)

Contradiction: COX-2 inhibition reduces PGE₂-mediated immunosuppression but may impair tissue repair and wound healing needed for anti-tumor responses.

Proposed Resolution: Targeted delivery to tumor microenvironment to achieve local COX-2 inhibition without systemic effects.

Outcome: ✗ **Rejected**—delivery optimization does not address the core mechanistic limitation that COX-2 inhibition has weak direct effects on T-cell exhaustion programs

53

54 S6 Tournament Judge Protocol and Prompts

55 The tournament system uses a Swiss-system structure with the following parameters: 3 rounds,
 56 maximum 50 pairwise comparisons, TrueSkill rating initialization ($\mu = 25$, $\sigma = 8.33$). Rankings
 57 are computed using conservative estimates ($\mu - 3\sigma$).

58 S6.1 Tournament Judge System Prompt

LLM Judge Prompt (GPT 5.1)

You are an expert biomedical scientist evaluating drug repurposing candidates for T-cell exhaustion. Compare two candidates and select the more promising one based on:

1. GNN Structural Evidence: Higher positive scores indicate stronger knowledge graph support for drug-disease association. Negative scores suggest the relationship is not well-captured by graph topology.
 2. Literature Support: Consider the number and quality of publications linking the drug to T-cell exhaustion or related immunomodulatory effects.
 3. Mechanistic Plausibility: Evaluate whether the proposed mechanism (checkpoint modulation, metabolic reprogramming, epigenetic remodeling, etc.) is biologically sound for reversing T-cell exhaustion.
- Output format: - Winner: [Candidate A/B] - Confidence: [High/Medium/Low] - Reasoning: [2-3 sentences explaining the decision]

59

60 S6.2 Agent Generation Prompts

61 Each specialized agent receives a domain-specific prompt:

- 62 • **Efficacy Agent:** “Generate 5 drug candidates targeting canonical T-cell exhaus-
63 tion pathways: PD-1/PD-L1 axis, TOX transcription factor, metabolic reprogramming
64 (AMPK/mTOR), or epigenetic regulators. Prioritize mechanistic directness.”
- 65 • **Safety Agent:** “Generate 5 drug candidates with established safety profiles suitable for
66 combination with checkpoint inhibitors. Avoid candidates with immunosuppressive effects
67 that could worsen T-cell dysfunction.”
- 68 • **Novelty Agent:** “Generate 5 drug candidates from underexplored mechanisms: indirect
69 pathway modulators (kinase inhibitors, metabolic modulators), epigenetic agents, or cos-
70 stimulatory agonists. Prioritize candidates with <10 PubMed abstracts linking them to T-
71 cell exhaustion.”

72 S7 Tournament Matchup Examples

73 S7.1 Key Match: Pembrolizumab vs Rapamycin

Tournament Match

Candidate A: Pembrolizumab (GNN: +0.102, Lit: 0.54, 40 articles)

Candidate B: Rapamycin (GNN: -0.048, Lit: 0.44, 20 articles)

Winner: Pembrolizumab

Judge’s Reasoning: Pembrolizumab has a positive GNN score (+0.102) and stronger literature support (0.54 vs 0.44). As a PD-1 checkpoint inhibitor, it directly targets the canonical pathway for T-cell exhaustion reversal. While Rapamycin’s mTOR inhibition is mechanistically sound for promoting memory T-cell formation, the direct checkpoint blockade mechanism is more established for exhaustion reversal.

74

75 S7.2 Round 2: Ibrutinib vs Tofacitinib

Tournament Match

Candidate A: Tofacitinib (JAK1/3 inhibitor)

Candidate B: Ibrutinib (BTK/ITK inhibitor)

Winner: Ibrutinib

Judge’s Reasoning: Candidate B (Ibrutinib) has weaker KG score but substantial PubMed literature support specifically linking it to T-cell exhaustion biology, which is prioritized by the rules. Both mechanisms are plausible, but the real-world evidence base for Ibrutinib makes it more promising overall.

76

77 S7.3 Round 2: Dasatinib vs Apremilast

Tournament Match

Candidate A: Apremilast (PDE4 inhibitor)

Candidate B: Dasatinib (multi-kinase inhibitor)

Winner: Dasatinib

Judge’s Reasoning: Dasatinib has clearly stronger literature support (3 PubMed articles, 0.201) specifically linking it to T-cell exhaustion modulation, and a well-documented, reversible “resting” effect on T cells that is directly relevant to the target biology.

78

79 S8 LLM-Based Mechanistic Evaluation Ratings

Table S7: LLM-based mechanistic evaluation ratings for final candidates (GPT 5.1). **Novelty** column reflects literature verification status (January 2026).

Drug	Rating	Novelty	Assessment
<i>Known Therapeutics (Final Tournament Rankings)</i>			
Pembrolizumab	5/5	None	Standard-of-care (Rank #1); validates methodology
Rapamycin	5/5	None	mTOR inhibition for memory T-cell formation (Rank #2)
Decitabine	4/5	None	Positive control (Rank #3); literature since 2017 [5]
Lenalidomide	4/5	None	Positive control (Rank #4); literature since 2014 [6]
Metformin	4/5	None	AMPK activation for metabolic fitness (Rank #5)
Nivolumab	5/5	None	Checkpoint inhibitor (Rank #6); validates methodology
Ibrutinib	4/5	None	Positive control (Rank #7); literature since 2013 [7]
<i>Novel Candidates (from separate GNN query)</i>			
Arbaclofen	4/5	HIGH	NOVEL: GABA-B/CXCR4 modulation; 0 exhaustion abstracts
Dotinurad	4/5	HIGH	NOVEL: URAT1/uric acid axis; 0 exhaustion abstracts
Vofopitant	4/5	HIGH	NOVEL: NK1R/Substance P blockade; 0 exhaustion abstracts

80 S9 Full LLM Mechanistic Evaluations

81 The following LLM-generated assessments (GPT 5.1) provide detailed mechanistic analysis, evi-
 82 dence quality evaluation, and clinical applicability ratings for each candidate. Ratings reflect repur-
 83 posing potential specifically for T-cell exhaustion modulation.

Key Finding: Negative GNN Score Yet Rank #2

Literature Status: ESTABLISHED (Positive Control)

- Extensive literature on mTOR inhibition and T-cell memory formation
- Araki et al. [3]: mTOR inhibition promotes memory CD8+ T-cell formation
- Pollizzi & Powell [4]: Comprehensive review of mTOR in T-cell differentiation
- Gabriel et al. [8]: mTORC1/mTORC2 selective regulation of CD8+ fates

Mechanistic Assessment

- mTOR inhibitor (FDA-approved for transplant rejection, certain cancers)
- **Key mechanism:** mTOR inhibition promotes memory T-cell formation over effector differentiation
- Low-dose rapamycin prevents terminal exhaustion by favoring memory-like progenitor states
- mTORC1 inhibition enhances fatty acid oxidation (FAO) for long-term T-cell survival
- **Mechanistically distinct** from checkpoint blockade—addresses metabolic/differentiation trajectory

Evidence Quality

- GNN score: -0.048 (31st percentile) → **negative despite strong literature**
- This demonstrates our key finding: LLM reasoning elevates mechanistically sound candidates
- 20 PubMed articles linking rapamycin/mTOR to T-cell memory and exhaustion

Clinical Applicability

- FDA-approved with decades of safety data (immunosuppression at high doses)
- Dose-dependent effects: low doses promote memory, high doses suppress
- Clear validation path: ex vivo treatment of exhausted T cells, measure memory markers (TCF-1, CD127)
- Potential combination with checkpoint inhibitors to maintain reinvigorated populations

Classification: Rank #2 despite negative GNN—demonstrates LLM reasoning value

86 **S9.2 Nivolumab (★★★★★) — Rank #6**

Standard-of-Care Checkpoint Inhibitor

Mechanistic Assessment

- Anti-PD-1 monoclonal antibody (FDA-approved for multiple cancers)
- Directly blocks PD-1/PD-L1 interaction, restoring T-cell function
- First-in-class PD-1 inhibitor [9]
- Mechanism identical to pembrolizumab; validates methodology

Evidence Quality

- GNN score: +0.065 (74.8th percentile) → moderate-strong structural support
- Literature: 37 articles, score 0.53 → very strong evidence base
- Standard-of-care; not a novel repurposing but validates methodology

Classification: Standard-of-care checkpoint inhibitor; validates methodology

87

88 **S9.3 Decitabine (★★★★) — Rank #3**

Positive Control: Established Literature Since 2017

Literature Status: NOT NOVEL

- Ghoneim et al. [5]: Dnmt3a-mediated exhaustion epigenetics (Cell 2017)
- Wang et al. [10]: Epigenetic reprogramming of exhausted CAR-T cells
- You et al. [11]: Decitabine enhances CAR-T persistence
- Zebley et al. [12]: DNA methylation dynamics in exhaustion (Nat Immunol 2021)
- **2017–present:** Extensive literature on epigenetic exhaustion reversal

Mechanistic Assessment

- DNA hypomethylating agent (FDA-approved for MDS, AML)
- Inhibits DNMT1/DNMT3a, reversing de novo methylation at exhaustion-associated loci
- Reopens chromatin at effector gene loci (IFN- γ , TNF- α , IL-2)
- Addresses the **epigenetic “fixation”** that limits checkpoint inhibitor efficacy
- Mechanism is **complementary to PD-1 blockade**—targets distinct barrier

Evidence Quality

- Literature: 10+ articles, score 0.35 → strong body of work on epigenetic exhaustion
- GNN score: +0.011 (55th percentile) → weak positive signal

Classification: Positive control (validates methodology; extensively studied since 2017)

89

90 **S9.4 Pembrolizumab (★★★★★) — Rank #1**

Standard-of-Care Checkpoint Inhibitor (Rank #1)

Mechanistic Assessment

- Anti-PD-1 monoclonal antibody; directly blocks PD-1/PD-L1 and reverses T-cell exhaustion
- PD-1 is a *defining* receptor of exhausted T cells
- Mechanism is **checkpoint blockade**—the canonical approach to reinvigorate exhausted T cells

Evidence Quality

- Literature: 40 articles, score 0.54 → very strong evidence base
- GNN score: +0.102 (87th percentile) → positive signal reflecting improved KG encoding
- Rank #1 validates methodology by placing standard-of-care at top

Clinical Applicability

- Approved across many solid tumors and some heme malignancies
- Used specifically to reinvigorate exhausted tumor-infiltrating T cells
- This is **standard-of-care**, not a novel repurposing

Classification: Standard-of-care; placing it at rank #1 validates our methodology

91

92 **S9.5 Ibrutinib (★★★★) — Rank #7**

Positive Control: Extensive Literature Since 2013

Literature Status: NOT NOVEL

- Dubovsky et al. [7]: ITK irreversible inhibitor first identified (Blood 2013)
- Sagiv-Barfi et al. [13]: Anti-PD-L1 + ibrutinib synergy (PNAS 2015)
- Li et al. [14]: BTK-independent exhaustion reversal directly proven (Front Immunol 2023)
- Papazoglou et al. [15]: CD8+ T-cell reinvigoration in E1912 trial (Blood 2024)
- **2013–present:** Extensively studied for T-cell exhaustion reversal

Mechanistic Assessment

- BTK inhibitor with off-target **ITK inhibition** in T cells
- Shifts signaling toward Th1 and away from Th2; enhances cytotoxic T-cell responses
- Alters tumor microenvironment (reduced suppressive cells; improved effector function)
- By promoting Th1/cytotoxic bias and improving TME, ibrutinib can **partially reverse exhausted-like phenotypes**

Evidence Quality

- Literature: 11+ articles, score 0.36 → strong body of data on T-cell modulation
- GNN score: −0.091 → negative (reflects KG encoding limitations)

Classification: Positive control (validates methodology; extensively studied since 2013)

93

Positive Control: Extensive Prior Literature

Literature Status: NOT NOVEL

- Görgün et al. [6]: PD-1↓, CTLA-4↓ on T cells
- Vø et al. [16]: CD28-deficient T-cell rescue via Notch signaling
- **2014–2022**: Extensive literature on immunomodulatory effects in T cells
- Correct identification validates methodology; does not represent novel discovery

Mechanistic Assessment

- IMiD; binds cereblon and promotes degradation of Ikaros/Aiolos
- Increases IL-2, co-stimulation, improved T-cell synapse formation, reduced Treg function
- By strengthening co-stimulation and effector functions, may **delay or partially counter exhaustion**

Clinical Applicability

- Approved for multiple myeloma, MDS, mantle cell lymphoma
- **CAR-T or tumor-specific T cells**: enhance early activation and synapse quality
- Needs careful dosing; chronic hyper-activation could accelerate terminal differentiation

Classification: Positive control (extensive prior literature since 2014)

Known Therapeutic: Metabolic Reprogramming

Literature Status: ESTABLISHED

- Extensive literature on AMPK activation and T-cell metabolism
- Eikawa et al.: Metformin enhances anti-tumor T-cell responses via AMPK
- Pearce et al.: AMPK activation promotes memory T-cell formation
- **2015–present:** Growing body of work on metabolic exhaustion reversal

Mechanistic Assessment

- AMPK activator (FDA-approved for type 2 diabetes)
- **Key mechanism:** AMPK activation restores mitochondrial fitness
- Promotes oxidative phosphorylation over glycolysis in T cells
- Enhances fatty acid oxidation for long-term T-cell survival
- Addresses **metabolic dysfunction** underlying exhaustion

Evidence Quality

- GNN score: -0.021 (45th percentile) → weak negative signal
- Literature: 9 PubMed articles linking metformin to T-cell exhaustion/metabolism
- Demonstrates LLM reasoning elevates metabolically relevant candidates

Clinical Applicability

- FDA-approved with decades of safety data
- Well-tolerated, low cost, oral administration
- Clear validation path: ex vivo treatment of exhausted T cells, measure OXPHOS markers
- Potential combination with checkpoint inhibitors for metabolic support

Classification: Known therapeutic for T-cell metabolic reprogramming (Rank #5)

Low Priority—Weak Mechanistic Support

Mechanistic Assessment

- Selective BCL-2 inhibitor; pro-apoptotic, mainly against BCL-2–dependent malignant cells
- BCL-2 is important for survival of many lymphocyte subsets
- Indiscriminate inhibition can reduce T-cell numbers and survival
- No presented evidence that venetoclax **beneficially reprograms** exhausted T cells

Evidence Quality

- Literature: 6 articles, score 0.28 → sparse and likely indirect references
- GNN score: +0.174 → highest positive in set (98.2nd percentile), strong structural support
- No explicit mechanistic citations provided

Clinical Applicability

- Approved for CLL and AML (often with HMAs or anti-CD20 antibodies)
- Known for cytopenias, infections, and immunosuppression
- Not suitable as a pro-T-cell-fitness agent

Flags: Poor mechanistic justification for exhaustion; candidate rationale appears speculative

99

100 **S9.9 Novel Candidates from Non-Immunological Indications**

101 The following candidates were identified through a separate query for high-GNN-score drugs with
 102 **zero prior T-cell exhaustion literature**. These represent truly novel repurposing hypotheses from
 103 unrelated therapeutic areas.

NOVEL: GABA-B/CXCR4 Axis

Literature Status: NOVEL

- **Zero** PubMed abstracts linking Arbaclofen to T-cell exhaustion (verified January 2026)
- Primary indication: Autism spectrum disorder, spasticity [17]
- No prior exploration of neuro-immune crosstalk for exhaustion

Mechanistic Assessment

- GABA-B receptor agonist (R-enantiomer of baclofen)
- **Novel hypothesis:** GABA-B receptors allosterically modulate CXCR4 [18]
- CXCR4 promotes T-cell retention in immunosuppressive tumor niches
- Blocking CXCR4-mediated retention could release exhausted T cells from suppressive microenvironments

Evidence Quality

- GNN score: +0.149 (94.7th percentile) → strong structural support
- Represents unexplored neuro-immune axis for exhaustion

TRIZ Separation Analysis

- **SPACE:** SEPARATE—Arbaclofen is positioned in CNS/behavioral pharmacology with no substantive immunology work
- **TIME:** NO_RECENT—No 2020+ papers on Arbaclofen + T-cell function
- **CONDITION:** Novel hypothesis—GABA-B receptor activation may allosterically modulate CXCR4, affecting T-cell retention in immunosuppressive niches

Inventive Level: HIGH (cross-domain, no prior art)

Philosophical Consensus: NOVEL (Hegel: NOVEL, Lakatos: NOVEL, TRIZ: NOVEL)

Classification: NOVEL—first candidate proposing GABA-B/CXCR4 modulation for T-cell exhaustion

NOVEL: Uric Acid/T-Cell Axis

Literature Status: NOVEL

- **Zero** PubMed abstracts linking Dotinurad to T-cell exhaustion (verified January 2026)
- Primary indication: Gout (URAT1 inhibitor, Japan) [19]
- No prior exploration of urate metabolism for T-cell exhaustion

Mechanistic Assessment

- Selective URAT1 (urate transporter 1) inhibitor
- **Novel hypothesis:** Elevated uric acid dysregulates T-cell function via NLRP3 [20]
- Chronic hyperuricemia creates pro-inflammatory milieu driving exhaustion
- URAT1 inhibition could reduce uric acid-mediated T-cell dysfunction

Evidence Quality

- GNN score: +0.165 (96.4th percentile) → very strong structural support
- Metabolic approach from unexpected direction (hyperuricemia)

TRIZ Separation Analysis

- **SPACE:** SEPARATE—Dotinurad is firmly in metabolic/gout/renal pharmacology, not immunology
- **TIME:** NO_RECENT—No papers on Dotinurad + immune modulation
- **CONDITION:** Novel hypothesis—URAT1 inhibition reduces uric acid, potentially dampening NLRP3 inflammasome activation and chronic inflammatory signals that drive exhaustion

Inventive Level: HIGH (metabolic → adaptive immunology)

Philosophical Consensus: NOVEL (Hegel: NOVEL, Lakatos: NOVEL, TRIZ: NOVEL)

Classification: NOVEL—first candidate proposing uric acid modulation for T-cell exhaustion

NOVEL: NK1R/Substance P Axis

Literature Status: NOVEL

- **Zero** PubMed abstracts linking Vofopitant to T-cell exhaustion (verified January 2026)
- Primary indication: Anxiety, PTSD (NK1 receptor antagonist) [21]
- No prior exploration of Substance P blockade for exhaustion

Mechanistic Assessment

- NK1 receptor (neurokinin-1) antagonist
- **Novel hypothesis:** Substance P promotes neurogenic inflammation [22]
- NK1 signaling has immunomodulatory effects on T cells
- Blocking Substance P could reduce chronic inflammatory stimulation driving exhaustion

Evidence Quality

- GNN score: +0.159 (95.9th percentile) → very strong structural support
- Represents unexplored neuro-immune axis (Substance P)

TRIZ Separation Analysis

- **SPACE:** SEPARATE—Vofopitant is in neuropsychopharmacology, not immunology
- **TIME:** NO_RECENT—No papers on Vofopitant + immune modulation
- **CONDITION:** Novel hypothesis—NK1 blockade disrupts Substance P neuro-immune signaling that may sustain exhausted T-cell phenotypes in chronic inflammation

Inventive Level: HIGH (psychiatry → T-cell immunotherapy)

Philosophical Consensus: NOVEL (Hegel: NOVEL, Lakatos: NOVEL, TRIZ: NOVEL)

Classification: NOVEL—first candidate proposing NK1R antagonism for T-cell exhaustion

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110 **S10 Meta-Review Synthesis**

111 The meta-review synthesizes patterns across all candidates to identify strategic directions and gaps
112 for future investigation.

Pattern Analysis**1. PD-1 Axis is the Central Pillar**

- **Pembrolizumab/Nivolumab:** Direct PD-1 blockade (ranks #1 and #6)
- **Rapamycin:** mTOR inhibition promotes memory T-cell formation (rank #2)
- **Decitabine:** Epigenetic reversal complements checkpoint blockade (rank #3)
- Top candidates converge on **canonical exhaustion reversal mechanisms**

2. T-Cell Metabolic and Epigenetic Fitness

- **Rapamycin:** mTOR↓, FAO↑, memory-like progenitor states
- **Decitabine:** DNMT inhibition reopens silenced effector loci
- **PD-1 blockers:** Restore anabolic metabolism once inhibitory signaling removed
- Strong theme of **metabolic and epigenetic “rejuvenation”** of exhausted T cells

3. Novel Hypotheses from Unexpected Directions

- **Arbaclofen:** GABA-B/CXCR4 neuro-immune axis
- **Dotinurad:** Uric acid/NLRP3 metabolic axis
- **Vofopitant:** Substance P/NK1R neuro-immune axis
- High GNN scores + zero exhaustion literature = truly novel repurposing candidates

4. Combination-Ready Design

- **Backbone:** PD-1 blockade (Pembrolizumab/Nivolumab)
- **Add-ons:** Rapamycin (metabolic), Decitabine (epigenetic), novel candidates
- Implicit strategy: **“Checkpoint anchor + metabolic + epigenetic modulators”**

115 S10.2 Identified Gaps and Underrepresented Approaches

Gap Analysis

1. No Direct Epigenetic Programming of Exhaustion

- Missing: BET inhibitors, HDAC inhibitors, DNMT inhibitors for exhaustion-specific chromatin
- Missing: Agents targeting core exhaustion TFs (TOX, NR4A family)
- Gap: **Intrinsic transcriptional/epigenetic locking of Tex** not addressed

2. Limited Checkpoint Breadth

- Only PD-1 is directly targeted
- Missing: TIM-3, LAG-3, TIGIT, CTLA-4, VISTA/BTLA modulators
- Gap: **Single-checkpoint focus** when persistent exhaustion involves multiple inhibitory receptors

3. Thin Co-stimulation and γ_c Cytokine Coverage

- Missing: Direct agonists of CD28, 4-1BB, OX40, ICOS, CD27
- Missing: Cytokine strategies (IL-7, IL-15, IL-21, improved IL-2 variants)
- Gap: Lack of **precise, tunable co-stimulatory interventions**

4. Incomplete TME Immunosuppressive Pathway Coverage

- Missing: Adenosine (A2A/A2B) pathway inhibitors
- Missing: IDO1/TDO inhibitors, TGF- β pathway modulators
- Missing: Chemokine axis modulation (CXCR3, CCR5)
- Gap: Many **extrinsic immunosuppressive circuits** not represented

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117 S10.3 Overarching Strategy

Strategic Framework

The candidate set implicitly endorses a **multi-axis exhaustion control strategy**:

Use **PD-1 blockade as the core**, then layer:

1. **Metabolic support** (favor memory-like and progenitor exhausted states)
2. **Co-stimulatory enhancement and IL-2 support**
3. **Microenvironmental debulking** (anti-VEGF/MDSC)

to sustain functional, less terminally exhausted T cells under chronic antigen exposure.

Next Iteration Priorities:

1. Treat PD-1 as fixed backbone; focus new hypotheses on complementary mechanisms
2. Add non-PD-1 checkpoint modulators (TIM-3, LAG-3, TIGIT)
3. Introduce epigenetic reprogrammers tied to Tex biology
4. Systematically explore γ_c cytokine and co-stimulatory strategies
5. Broaden TME modulation beyond VEGF (adenosine, IDO, TGF- β)

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Table S8: Mechanistic rationales and clinical contexts for all final candidates.

Drug	Mechanism	Clinical Context	Key Evidence
<i>Known Therapeutics (Final Tournament Rankings)</i>			
Pembrolizumab	Direct PD-1 blockade; reverses checkpoint-mediated T-cell exhaustion; canonical mechanism	Multiple solid tumors and hematologic malignancies; standard-of-care	GNN +0.102; Lit 0.54; Rank #1
Rapamycin	mTOR inhibition promotes memory T-cell formation; prevents terminal differentiation; FAO enhancement	Combination with checkpoint inhibitors; low-dose regimens	GNN −0.048; Lit 0.44; Rank #2
Decitabine	DNA hypomethylation; reverses exhaustion-specific epigenetic programs; reopens effector gene loci	MDS/AML; IO combinations; CAR-T enhancement	GNN +0.011; Lit 0.35; Rank #3
Lenalidomide	Cereblon binding; Ikaros/Aiolos degradation; enhanced co-stimulation	CAR-T persistence; vaccine enhancement	Lit 0.38; Rank #4 [6]
Metformin	AMPK activation restores mitochondrial fitness and oxidative metabolism	Diabetes; cancer adjuvant; metabolic reprogramming	GNN −0.021; Lit 0.33; Rank #5
Nivolumab	Anti-PD-1 monoclonal antibody; reinvigorates exhausted T cells in chronic antigen settings	Solid tumors, lymphomas; directly reverses tumor T-cell exhaustion	GNN +0.065; Lit 0.53; Rank #6
Ibrutinib	BTK/ITK inhibition; shifts Th1/cytotoxic bias; improves TME via reduced suppressive cells	Hematologic + checkpoint combinations	Lit 0.36; Rank #7 [7]
<i>Novel Candidates (from separate GNN query)</i>			
Arbaclofen	GABA-B agonist; CXCR4 allosteric modulation; releases T cells from suppressive niches	Autism/spasticity (primary); novel T-cell hypothesis	GNN +0.149; 0 exhaustion articles
Dotinurad	URAT1 inhibitor; reduces uric acid-mediated NLRP3 activation and T-cell dysfunction	Gout (primary); novel metabolic hypothesis	GNN +0.165; 0 exhaustion articles
Vofopitant	NK1R antagonist; blocks Substance P neurogenic inflammation; neuro-immune axis	Anxiety/PTSD (primary); novel neuro-immune hypothesis	GNN +0.159; 0 exhaustion articles

S11 Complete Candidate Rationales

S12 Glossary of Terms and Abbreviations

S13 Temporal Generalization (Zero-Shot) Validation

To assess the framework’s reasoning capabilities beyond its training data cutoff, we conducted a “temporal holdout” validation. We selected three indications where major repurposing approvals or key clinical findings occurred *after* September 2024 (the approximate knowledge cutoff for LLMs used in this study).

Analysis: While the system did not identify the exact novel agents (Tirzepatide, Semaglutide), which were sparsely represented in the pre-2024 knowledge graph, it successfully recovered the **correct therapeutic drug class** in 67% (2/3) of cases. For OSA, the model identified Liraglutide (46 mentions across evolution rounds)—the same GLP-1 agonist class as Tirzepatide. For CKD, it identified both Liraglutide (18 mentions) and SGLT2 inhibitors (237 mentions), capturing the mechanistic rationale for metabolic and hemodynamic renoprotection.

The failure case (AUD) is instructive: Semaglutide’s effect on alcohol consumption through GLP-1 receptor modulation of reward pathways represents a *genuinely novel mechanism* that emerged from

Table S9: Glossary of abbreviations and technical terms used in this paper.

Term	Definition
<i>Evaluation Metrics</i>	
AUROC	Area Under the Receiver Operating Characteristic Curve; measures discriminative ability across all classification thresholds (0–1, higher is better)
AUPRC	Area Under the Precision-Recall Curve; measures precision-recall trade-off, particularly useful for imbalanced datasets
MRR	Mean Reciprocal Rank; average of $1/\text{rank}$ across a set of queries; higher indicates better ranking quality
Hits@K	Proportion of known positives appearing in top-K ranked positions
TrueSkill	Bayesian skill rating system [23] modeling quality as Gaussian (μ =mean, σ =uncertainty); we rank by $\mu - 3\sigma$
<i>Biological Terms</i>	
AMPK	AMP-activated Protein Kinase; master metabolic regulator activated by low energy states
BCL-2	B-cell Lymphoma 2; anti-apoptotic protein regulating cell survival
BCR	B-cell Receptor; antigen recognition complex on B lymphocytes
BTK	Bruton’s Tyrosine Kinase; kinase in BCR signaling pathway
CAR-T	Chimeric Antigen Receptor T-cell; engineered T cells expressing synthetic receptors
CTLA-4	Cytotoxic T-Lymphocyte-Associated protein 4; inhibitory checkpoint receptor
ITK	Interleukin-2-inducible T-cell Kinase; kinase in TCR signaling pathway
LAG-3	Lymphocyte-Activation Gene 3; inhibitory checkpoint receptor
mTOR	Mechanistic Target of Rapamycin; central growth/metabolism regulator
NR4A	Nuclear Receptor subfamily 4 group A; transcription factors associated with exhaustion
OXPHOS	Oxidative Phosphorylation; mitochondrial ATP production pathway
PD-1	Programmed cell Death protein 1; key inhibitory checkpoint receptor on exhausted T cells
PD-L1	Programmed Death-Ligand 1; ligand for PD-1 expressed on tumor/stromal cells
TCF-1/TCF7	T Cell Factor 1 (gene: TCF7); transcription factor marking stem-like T cells
TIM-3	T-cell Immunoglobulin and Mucin-domain containing-3; inhibitory checkpoint receptor
TME	Tumor Microenvironment; cellular/molecular context surrounding tumor cells
TOX	Thymocyte Selection Associated HMG Box; transcription factor driving exhaustion program
<i>Framework Terms</i>	
GNN Score	Predicted link probability from knowledge graph neural network, scaled to $[-1, +1]$; positive indicates structural support
Lit. Score	Literature Support score from PubMed abstract analysis, normalized to $[0, 1]$
SOC	Standard-of-Care; established first-line treatment for a condition
W/L/T	Wins/Losses/Ties in tournament pairwise comparisons
<i>Philosophical Frameworks</i>	
Hegelian	Dialectical method: Thesis $\rightarrow$ Antithesis $\rightarrow$ Synthesis for hypothesis evolution
Lakatosian	Research programme methodology: immutable “hard core” + adjustable “protective belt”
TRIZ	Theory of Inventive Problem Solving; systematic contradiction resolution from engineering

Table S10: Temporal holdout validation on indications with post-September 2024 approvals. **Target Drug:** The ground-truth drug approved/validated after the model’s knowledge cutoff. **Predicted Drug:** The model’s top-ranked candidate(s) from the same mechanistic class. **Result:** Whether the model identified the correct therapeutic class despite lacking knowledge of the specific approval.

Indication	Target Drug (Post-2024)	Predicted Drug	Result
Obstructive Sleep Apnea (OSA)	Tirzepatide (GLP-1/GIP agonist; FDA Dec 2024)	Liraglutide (GLP-1 agonist)	✓ Class Match: Identified GLP-1 agonist for metabolic/weight-related mechanism
Chronic Kidney Disease (CKD)	Semaglutide (GLP-1 agonist; FDA Jan 2025)	Liraglutide + SGLT2 inhibitors	✓ Class Match: Identified both GLP-1 and complementary renoprotective mechanisms
Alcohol Use Disorder (AUD)	Semaglutide (GLP-1 agonist; JAMA Psychiatry Feb 2025)	Naltrexone, Topiramate, Baclofen	✗ Miss: Failed to identify GLP-1 reward pathway modulation—a genuinely novel mechanism

unexpected clinical observations rather than predictable pharmacology. This suggests the methodology captures established mechanistic relationships but cannot predict truly novel mechanism-of-action discoveries—an honest limitation of any knowledge-graph-grounded approach.

Implications: This temporal holdout validation provides evidence that the hybrid GNN+LLM framework performs mechanistic reasoning rather than memorizing associations from training data. The ability to identify the correct drug class (GLP-1 agonists) for OSA and CKD, despite lacking knowledge of the specific post-2024 approvals, suggests the methodology can generalize to novel contexts within established mechanistic frameworks.

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